

AMS 580 Quiz 2

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AMS 580 Quiz Logistic Regression

Library

```
library(MASS)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.0
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## x dplyr::select() masks MASS::select()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(leaps)
```

```
## Warning: package 'leaps' was built under R version 4.3.3
```

```
library(car)
```

```
## Warning: package 'car' was built under R version 4.3.3
```

```
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##     recode
##
## The following object is masked from 'package:purrr':
##
##     some
```

```
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.3.3
```

```
## Loading required package: lattice
```

```
##
```

```
## Attaching package: 'caret'
```

```
##
```

```
## The following object is masked from 'package:purrr':
```

```
##
```

```
## lift
```

```
library(dplyr)
```

```
library(lmtest)
```

```
## Warning: package 'lmtest' was built under R version 4.3.3
```

```
## Loading required package: zoo
```

```
## Warning: package 'zoo' was built under R version 4.3.3
```

```
##
```

```
## Attaching package: 'zoo'
```

```
##
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## as.Date, as.Date.numeric
```

Data import

```
titantic <- read.table("/Users/thomashuang/Downloads/Titanic2.csv", sep = ",", header=TRUE)
head(titantic)
```

```
## PassengerId Survived Pclass
```

```
## 1 1 0 3
```

```
## 2 2 1 1
```

```
## 3 3 1 3
```

```
## 4 4 1 1
```

```
## 5 5 0 3
```

```
## 6 6 0 3
```

```
## Name Sex Age SibSp Parch
```

```
## 1 Braund, Mr. Owen Harris male 22 1 0
```

```
## 2 Cumings, Mrs. John Bradley (Florence Briggs Thayer) female 38 1 0
```

```
## 3 Heikkinen, Miss. Laina female 26 0 0
```

```
## 4 Futrelle, Mrs. Jacques Heath (Lily May Peel) female 35 1 0
```

```
## 5 Allen, Mr. William Henry male 35 0 0
```

```
## 6 Moran, Mr. James male NA 0 0
```

```
## Ticket Fare Cabin Embarked
```

```
## 1 A/5 21171 7.2500 S
```

```
## 2      PC 17599 71.2833  C85      C
## 3 STON/O2. 3101282  7.9250      S
## 4      113803 53.1000  C123      S
## 5      373450  8.0500      S
## 6      330877  8.4583      Q
```

```
str(titanic)
```

```
## 'data.frame':  891 obs. of  12 variables:
## $ PassengerId: int  1 2 3 4 5 6 7 8 9 10 ...
## $ Survived   : int  0 1 1 1 0 0 0 0 1 1 ...
## $ Pclass     : int  3 1 3 1 3 3 1 3 3 2 ...
## $ Name       : chr  "Braund, Mr. Owen Harris" "Cumings, Mrs. John Bradley (Florence Briggs Thayer)"
## $ Sex        : chr  "male" "female" "female" "female" ...
## $ Age        : num  22 38 26 35 35 NA 54 2 27 14 ...
## $ SibSp      : int  1 1 0 1 0 0 0 3 0 1 ...
## $ Parch      : int  0 0 0 0 0 0 0 1 2 0 ...
## $ Ticket     : chr  "A/5 21171" "PC 17599" "STON/O2. 3101282" "113803" ...
## $ Fare       : num  7.25 71.28 7.92 53.1 8.05 ...
## $ Cabin      : chr  "" "C85" "" "C123" ...
## $ Embarked   : chr  "S" "C" "S" "S" ...
```

Data pre-processing

```
#exclude particular columns
titanic <- subset(titanic, select = -c(PassengerId,Name,Ticket,Cabin))
str(titanic)
```

```
## 'data.frame':  891 obs. of  8 variables:
## $ Survived: int  0 1 1 1 0 0 0 0 1 1 ...
## $ Pclass  : int  3 1 3 1 3 3 1 3 3 2 ...
## $ Sex     : chr  "male" "female" "female" "female" ...
## $ Age     : num  22 38 26 35 35 NA 54 2 27 14 ...
## $ SibSp   : int  1 1 0 1 0 0 0 3 0 1 ...
## $ Parch   : int  0 0 0 0 0 0 0 1 2 0 ...
## $ Fare    : num  7.25 71.28 7.92 53.1 8.05 ...
## $ Embarked: chr  "S" "C" "S" "S" ...
```

```
#omit missing values
titanic <- na.omit(titanic)
str(titanic)
```

```
## 'data.frame':  714 obs. of  8 variables:
## $ Survived: int  0 1 1 1 0 0 0 1 1 1 ...
## $ Pclass  : int  3 1 3 1 3 1 3 3 2 3 ...
## $ Sex     : chr  "male" "female" "female" "female" ...
## $ Age     : num  22 38 26 35 35 54 2 27 14 4 ...
## $ SibSp   : int  1 1 0 1 0 0 3 0 1 1 ...
## $ Parch   : int  0 0 0 0 0 0 1 2 0 1 ...
## $ Fare    : num  7.25 71.28 7.92 53.1 8.05 ...
```

```
## $ Embarked: chr "S" "C" "S" "S" ...
## - attr(*, "na.action")= 'omit' Named int [1:177] 6 18 20 27 29 30 32 33 37 43 ...
## ..- attr(*, "names")= chr [1:177] "6" "18" "20" "27" ...
```

```
#factorize survived and Pclass column
titantic$Survived <- factor(titantic$Survived)
titantic$Pclass <- factor(titantic$Pclass)
str(titantic)
```

```
## 'data.frame': 714 obs. of 8 variables:
## $ Survived: Factor w/ 2 levels "0","1": 1 2 2 2 1 1 1 2 2 2 ...
## $ Pclass : Factor w/ 3 levels "1","2","3": 3 1 3 1 3 1 3 3 2 3 ...
## $ Sex : chr "male" "female" "female" "female" ...
## $ Age : num 22 38 26 35 35 54 2 27 14 4 ...
## $ SibSp : int 1 1 0 1 0 0 3 0 1 1 ...
## $ Parch : int 0 0 0 0 0 0 1 2 0 1 ...
## $ Fare : num 7.25 71.28 7.92 53.1 8.05 ...
## $ Embarked: chr "S" "C" "S" "S" ...
## - attr(*, "na.action")= 'omit' Named int [1:177] 6 18 20 27 29 30 32 33 37 43 ...
## ..- attr(*, "names")= chr [1:177] "6" "18" "20" "27" ...
```

After cleaning, there are only 714 observations left.

Data Splitting

```
set.seed(123)
training.samples <- titantic$Survived %>% createDataPartition(p=0.8, list=FALSE)
train.data <- titantic[training.samples,]
test.data <- titantic[-training.samples,]
```

Model fitting

```
#fit the model
fit <- glm(Survived ~., data=train.data, family = binomial)

#summary
summary(fit)
```

```
##
## Call:
## glm(formula = Survived ~ ., family = binomial, data = train.data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  4.128749   0.604277   6.833 8.34e-12 ***
## Pclass2      -1.012074   0.370945  -2.728 0.006365 **
## Pclass3      -2.076496   0.384989  -5.394 6.90e-08 ***
## Sexmale      -2.556944   0.242699 -10.535 < 2e-16 ***
```

```
## Age          -0.031944    0.009200   -3.472 0.000516 ***
## SibSp        -0.350734    0.144212   -2.432 0.015012 *
## Parch        -0.115459    0.139457   -0.828 0.407717
## Fare         0.002723    0.003202    0.851 0.395005
## EmbarkedQ    -1.080480    0.641079   -1.685 0.091910 .
## EmbarkedS    -0.718802    0.320819   -2.241 0.025057 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 772.45  on 571  degrees of freedom
## Residual deviance: 518.19  on 562  degrees of freedom
## AIC: 538.19
##
## Number of Fisher Scoring iterations: 5
```

Confusion Matrix

```
# find Y hat of each x
pred_prob <- fit %>% predict(test.data, type="response")

#create predicted class
predicted.classes <- ifelse(pred_prob > 0.5, "survived", "dead")

test.data$Survived <- ifelse(test.data$Survived == 1, "survived", "dead")

#confusion matrix
table(predicted.classes, test.data$Survived)
```

Manual Calculation

```
##
## predicted.classes dead survived
##      dead      71      15
##      survived  13      43
```

```
cm <- confusionMatrix(factor(predicted.classes), factor(test.data$Survived), positive = "survived")
```

Built-in Function When positive class = “1” Detect survived

Actual/Predicted	Negative (Actual 0)	Positive (Actual 1)
Negative (Predicted 0)	**True Negative (TN)**	**False Negative (FN)**
Positive (Predicted 1)	**False Positive (FP)**	**True Positive (TP)**

Overall accuracy

$$P(\text{Predicted Correctly}) = \frac{TP+TN}{TP+TN+FP+FN}$$

```
(71+43)/(71+15+13+43)
```

```
## [1] 0.8028169
```

```
cm$overall["Accuracy"]
```

```
## Accuracy  
## 0.8028169
```

Sensitivity

$$\text{Sensitivity} = P(\text{true positive}|\text{correct}) = \frac{TP}{TP+FN}$$

```
(43)/(43+15)
```

```
## [1] 0.7413793
```

```
cm$byClass["Sensitivity"]
```

```
## Sensitivity  
## 0.7413793
```

Specificity

$$\text{Specificity} = P(\text{true negative}|\text{correct}) = \frac{TN}{TN+FP}$$

```
(71)/(13+71)
```

```
## [1] 0.8452381
```

```
cm$byClass["Specificity"]
```

```
## Specificity  
## 0.8452381
```

Prediction

```
new_data <- data.frame(  
  Pclass = factor(c(3,1,2)),  
  Sex = c("male","female","male"),  
  Age = c(24,68,41),  
  SibSp = c(1,0,1),  
  Parch = c(0,0,2),  
  Fare = c(8.42,24.34,41.93),  
  Embarked=c("Q","C","S")  
)  
new_data
```

```
##      Pclass      Sex Age SibSp Parch  Fare Embarked
## 1         3   male  24     1     0   8.42         Q
## 2         1 female  68     0     0  24.34         C
## 3         2   male  41     1     2  41.93         S
```

```
# find  $\hat{Y}$  of a given  $X$ 
pred_fit <- predict(fit, new_data, type="response")
pred_fit
```

```
##           1           2           3
## 0.06418536 0.88317148 0.12606024
```

```
result <- ifelse(pred_fit > 0.5, "survived", "dead")
result
```

```
##           1           2           3
## "dead" "survived" "dead"
```

- Passenger ID 892 will not survived.
- Passenger ID 893 will survived.
- Passenger ID 894 will no survived